IT3091 Machine Learning - Personal Learning Journey

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My main responsibility was the weekly price-model comparison and evaluation stage. I learned that the target definition must stay connected to the business question: our price output is the next observed weekly wholesale price for the stated Colombo market, not a national price estimate.

The price pipeline compared a dummy baseline with Linear Regression, Random Forest, and Gradient Boosting. The records through 2019 were used for training, and the later 2020-onward period contained 6,159 evaluation rows. Linear Regression had the best reported results in that comparison, with MAE around Rs 27.24/kg, RMSE around Rs 48.33/kg, and R-squared around 0.906.

I learned to explain these as regression metrics, not a percentage of predictions that are correct. MAE and RMSE are measured in rupees per kilogram, while R-squared describes variation captured relative to a reference. The evaluation summarizes this historical split; it does not guarantee the next forecast or prove performance in every future market period.

The Streamlit app uses the saved weekly model and asks for a crop, observation week, current Colombo wholesale price, and previous-week price. It shows the next-week estimate and expected movement. The app's separate yield screen is backed by a different Random Forest pipeline prepared in Member 1's crop-yield notebook; keeping the models separate made the distinct data sources, units, and user inputs easier to explain.

If I continued the project, I would make the price evaluation stronger with a rolling-origin test and compare against simple last-value and seasonal baselines on untouched future weeks. This work taught me to report the actual evaluation design, preserve the weekly horizon, and connect saved model artifacts to a cautious user-facing display.